

ATROPOS OBLIGATION DAG

Source Document 7 → Hybrid RESEARCH+IMPL + Track/WIRE

Tier-1 Priority: Complete Extirpation of AppSourceTemplate.kt & LanguageScaffold.kt

Maximum atomization · Zero artificial templates · Every atom independently verifiable · 2026-10-06

1. Extraction Rules (Locked)

Maximum atomization. Blueprint phases do not consume concrete atoms. Every feature, function, utility, invariant, gate, driver, and mathematical claim is decomposed until it cannot be divided further.

Hybrid format: Residual Obligation DAG v3 RESEARCH (R:) + IMPL (I:) density combined with HOE Track / dependsOn / WIRE discipline.

Research prompts force external discovery of superior implementations; they never restate Source Doc 7 sketches.

Impl notes name foreseeable failure modes and the cheapest correct path that preserves existing ATROPOS substrate (IsolatedWorktreeService, VerifiedCompletionGate, BoundedProcessRunner, TerritoryGrant, ArchitectureComplianceChecker, SpecGraphCompiler, DLOI).

Status tags: RESIDUAL | AMENDMENT-CANDIDATE | NEW-SUBSTRATE | 2050-NEURO-SYMBOLIC | SUPERSEDED-CAPABILITY.

WIRE may name classes newly introduced by Source Doc 7. Final section of this document is a mandatory Caller-Completeness Gate that must pass before any promotion.

Tier discipline: Tier-1 (this document) must reach verified completion before any subsequent tier is scheduled. Tier-1 = total purge of static templates + live Interface-First Wave factory that produces real applications.

2. Tier Structure

Tier-1 (this DAG): Extirpation of AppSourceTemplate.kt / LanguageScaffold.kt + full Wave 0–4 factory substrate + native toolchains + 2050 neuro-symbolic gates. Acceptance = real applications compile and pass VerifiedCompletionGate with zero template bytes.

Tier-2 (future): Long-horizon storage constitution, swarm hierarchy, thermal/battery pacing, exact-hash restart continuity, quota ledger.

Tier-3 (future): Full CLI/Unix sovereignty, PTY process hosting, shell completion engine, OSC-52 clipboard, anti-fake-green oracles at production scale.

Tier-4 (future): Cloud delivery drivers (Vercel/Netlify/Supabase) + packaging to bit-verified ELF/APK/Web.

3. Tier-1 Atoms — Template Extirpation & Core Factory Substrate

3.1 Deprecation & Routing Severance

SD7-T1-001+extirpate-AppSourceTemplate [NEW-SUBSTRATE]

Track: S (Shared Core)

dependsOn: []

R: Research the complete call graph of AppSourceTemplate.kt and every consumer that imports mainSource(), testSource(), genericMain() or any other string-literal emitter. Identify the minimal set of deletion points that leave zero residual template paths.

I: Hard-delete the file from the critical path. Remove every import and call site. Replace with a single compile-time assertion that fails if any template string literal matching the old signatures is present in the factory package. Foreseeable error: residual string constants surviving in test fixtures or comments; scan with ArchitectureComplianceChecker.

WIRE: AppFactoryRouter.kt (mutate plan() and runLocal()), ArchitectureComplianceChecker (new ban rule), DagExecutionService.kt

SD7-T1-002+extirpate-LanguageScaffold [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-001]

R: Research every LanguageScaffold entry point and language-specific string templates. Locate any dynamic dispatch that still routes to scaffold methods.

I: Hard-delete. All language-specific scaffolding is replaced by Wave-0 ContractLatticeSynthesizer output. Foreseeable error: language-kind switch statements that still call scaffold; convert them to throw UnsupportedOperationException until Wave-0 contracts exist.

WIRE: AppProjectGenerator.kt, AppFactoryRouter.kt

SD7-T1-003+sever-AppFactoryRouter [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-001, SD7-T1-002]

R: Research clean router patterns that completely detach intent parsing from code emission. Examine how modern agent factories (Aider, Codex, OpenCode) keep planning and synthesis orthogonal.

I: Mutate AppFactoryRouter so plan() and runLocal() emit only a typed WaveExecutionRequest and hand control to AutonomousWaveOrchestrator. No template path may remain reachable. Foreseeable error: fallback branch that still calls generateApp() with templates; delete the fallback.

WIRE: AutonomousWaveOrchestrator (new), AppFactoryRouter.kt

SD7-T1-004+mutate-DagExecutionService [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-003]

R: Research batched topological wave processors that respect territory isolation and worktree lifecycle. Prefer designs that never hold the entire atom set in a single context window.

I: Replace synchronous execution stubs with wave-batch processing. Every PROVIDER_CALL node must execute inside an IsolatedWorktreeService worktree. Foreseeable error: shared mutable state across partitions; enforce territory disjointness before dispatch.

WIRE: DagExecutionService.kt, IsolatedWorktreeService, PartitionClusterManager

3.2 Wave-0 Substrate & Toolchains

SD7-T1-010+NativeToolchainRegistry [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research deterministic toolchain discovery patterns used by Bazel, Pants, and Nix that locate compilers without hard-coded absolute paths and survive Termux / Linux / macOS differences.

I: Implement NativeToolchainRegistry that probes PATH and known locations, returns typed LanguageToolchainDriver instances, and never swallows non-zero exit codes. Foreseeable error: silent fallback to mock drivers; ban any driver that does not invoke a real binary.

WIRE: ToolchainExecutionDriver, BoundedProcessRunner, VerifiedCompletionGate

SD7-T1-011+LanguageToolchainDriver-interface [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010]

R: Research minimal interface contracts that cover compile, test, lint, package for heterogeneous languages while remaining extensible to new drivers.

I: Define LanguageToolchainDriver with compile(worktree), test(worktree), lint(worktree), package(worktree). All methods return structured Diagnostic maps (file, line, column, code, message). Foreseeable error: drivers that return success on empty worktrees; require at least one source file.

WIRE: NativeToolchainRegistry, all concrete drivers

SD7-T1-012+RustCargoDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research cargo check / cargo test / cargo build --release exit-code and diagnostic capture under bounded process runners.

I: Implement via BoundedProcessRunner. Capture stdout/stderr, map to Diagnostic. Non-zero exit = terminal failure. Foreseeable error: interactive prompts hanging the process; always supply --color=never and non-interactive flags.

WIRE: NativeToolchainRegistry, ToolchainExecutionDriver

SD7-T1-013+TypescriptNodeDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research pnpm/npm + tsc --noEmit + vite build diagnostic patterns and lockfile isolation.

I: Run inside isolated worktree. Use --frozen-lockfile when possible. Non-zero tsc or build exit fails the gate. Foreseeable error: node_modules leakage across worktrees; always install into the worktree only.

WIRE: NativeToolchainRegistry

SD7-T1-014+GoToolchainDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research go build / go vet / go test -race under resource limits.

I: BoundedProcessRunner with explicit GOMAXPROCS and timeout. Race detector enabled for test. Foreseeable error: module cache pollution; set GOPATH/GOCACHE inside worktree.

WIRE: NativeToolchainRegistry

SD7-T1-015+PythonFastApiDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research mypy --strict + ruff + pytest isolation patterns for FastAPI projects.

I: Create ephemeral venv inside worktree if needed. mypy --strict must pass before pytest. Foreseeable error: system site-packages contamination; always --isolated.

WIRE: NativeToolchainRegistry

SD7-T1-016+KotlinGradleDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011]

R: Research ./gradlew compileKotlin test assemble under daemon control and offline mode.

I: Prefer --offline when caches are present. Capture Gradle diagnostic XML if available. Non-zero exit fails. Foreseeable error: Gradle daemon leaking across worktrees; use --no-daemon or unique project cache.

WIRE: NativeToolchainRegistry, GovernedCompileGate

SD7-T1-017+AndroidSdkDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-011, SD7-T1-016]

R: Research AAPT2 + D8/R8 + gradlew assembleDebug production of unsigned APK with bit-level verification.

I: Produce real APK artifact. Verify APK exists and has non-zero size. Foreseeable error: missing ANDROID_HOME; fail closed with clear diagnostic rather than mock APK.

WIRE: NativeToolchainRegistry, CloudDeploymentAdapter (later tier)

SD7-T1-018+PartitionClusterManager [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research Kahn topological sort + greedy domain-path proximity clustering under token budgets. Examine how modern build systems (Bazel, Pants) partition for parallel execution while preserving acyclicity.

I: Implement Ψ operator. Constraints: quotient graph remains DAG, sum of tokens + interface context $\leq B_{\text{token}}$ (default 12 000), territories pairwise disjoint. Foreseeable error: cycles introduced by greedy merge; re-run topological check after every cluster formation.

WIRE: InternalExecutionDagSynthesizer, AutonomousWaveOrchestrator, TerritoryGrant

3.3 Wave-0 Contract Lattice

SD7-T1-020+ContractLatticeSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010, SD7-T1-018]

R: Research pure interface / trait / schema synthesis that emits zero executable statements. Examine OpenAPI 3.1, Protobuf, SQL DDL, TypeScript .d.ts, Rust traits, Kotlin interfaces for minimal complete contracts.

I: Wave-0 output contains only type signatures, schemas, and protocol definitions. Compilation gate is typecheck/lint only. Repair until clean. Foreseeable error: accidental inclusion of function bodies; AST walk rejects any executable node.

WIRE: AutonomousWaveOrchestrator, DecoupledWorkerSynthesizer, ArchitectureComplianceChecker

SD7-T1-021+clean-room-prompt-contract [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-020]

R: Research prompt formats that mathematically enforce $I(f;fk | I) = 0$ by construction. Examine context isolation techniques used in secure multi-party computation and clean-room design reviews.

I: AgentPromptContract must inject the Wave Contract format. Worker receives only atom payload + Wave-0 interfaces. Sibling implementation text is never present. Foreseeable error: accidental inclusion of previous worker diffs; scrub context before every dispatch.

WIRE: AgentPromptContract.kt, DecoupledWorkerSynthesizer

3.4 Wave-1 Behavioral Verification Suites

SD7-T1-030+Wave1-test-suite-generator [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-020, SD7-T1-021]

R: Research property-based and interface-mock test generation that must fail against empty stubs and compile against Wave-0 contracts only.

I: Generate isolated unit + property tests. Gate: tests compile against contracts and fail on empty implementations. Foreseeable error: tests that pass vacuously; require at least one failing assertion on a stub.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate

3.5 Wave-2 Parallel Partition Implementation

SD7-T1-040+DecoupledWorkerSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-018, SD7-T1-021, SD7-T1-030]

R: Research isolated worktree synthesis loops with bounded repair (max 3) that feed compiler diagnostics back without expanding context beyond the partition footprint.

I: Each partition runs in its own IsolatedWorktreeService worktree. Worker sees only its atoms + Wave-0 contracts. On failure, repairNode feeds stderr back. Merge only after clean verification. Foreseeable error: territory collision; fail closed and revert worktree.

WIRE: IsolatedWorktreeService, GovernedCompileGate, ProviderCascadeRouter, PartitionClusterManager

SD7-T1-041+CompilerDiagnosticRepairLoop [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-040]

R: Research diagnostic-to-prompt feedback loops that preserve the original atom and interface set while only mutating the failing file.

I: Max 3 attempts. Each attempt receives the exact compiler diagnostic map. No expansion of context. Foreseeable error: infinite repair oscillation; after 3 failures create a directed RepairNode and halt the partition.

WIRE: DecoupledWorkerSynthesizer, GovernedCompileGate

3.6 Wave-3 DI & Composition

SD7-T1-050+DependencyInjectionAssembler [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-040]

R: Research pure DI registry synthesis (Koin, Dagger, NestJS modules, FastAPI Depends, Go wire) that never hard-codes sibling constructors.

I: Wire concrete implementations to Wave-0 interfaces only. Ban static singletons and service locators. Foreseeable error: direct new of sibling classes; ArchitectureComplianceChecker must reject.

WIRE: AutonomousWaveOrchestrator, ArchitectureComplianceChecker

3.7 Wave-4 Global Verification & Orchestration

SD7-T1-060+AutonomousWaveOrchestrator [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-003, SD7-T1-018, SD7-T1-020, SD7-T1-040, SD7-T1-050]

R: Research five-wave execution engines that treat each wave as a hard gate and never proceed on partial success.

I: Execute Wave 0 → 1 → 2 → 3 → 4. Any Halted report aborts the build with diagnostics. Success produces artifact path + optional deployment URL. Foreseeable error: soft failures that continue; every non-zero gate is terminal.

WIRE: AppFactoryRouter, PartitionClusterManager, ContractLatticeSynthesizer, DecoupledWorkerSynthesizer, DependencyInjectionAssembler, NativeToolchainRegistry

SD7-T1-061+VerifiedCompletionGate-toolchain-binding [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010, SD7-T1-060]

R: Research binding of language-specific native exit codes directly into completion calculus so that fake success is impossible.

I: Mutate VerifiedCompletionGate to invoke the concrete LanguageToolchainDriver and treat any non-zero exit as terminal rejection. Foreseeable error: shell scripts that echo PASS; ban any verification that does not execute a real compiler binary.

WIRE: VerifiedCompletionGate.kt, NativeToolchainRegistry, BoundedProcessRunner

SD7-T1-062+mutate-ExecutionPolicyEngine [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-010]

R: Research policy engines that allow automated toolchain invocation inside verified sandboxes without human tokens while still enforcing territory and secret boundaries.

I: Permit cargo, go, pnpm, clang, zig, gradlew inside BoundedProcessRunner sandboxes. Secrets stripped from env. Foreseeable error: secret leakage into subprocess; always clean env.

WIRE: ExecutionPolicyEngine.kt, BoundedProcessRunner, TerritoryGrant

SD7-T1-063+mutate-InternalExecutionDagSynthesizer [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-018]

R: Research partition mapping that groups unbounded atoms into territorial clusters while emitting formal input/output interface ports and boundary hashes.

I: Implement Ψ . Emit partition metrics and handoff manifests. Foreseeable error: token budget overflow; reject partitions that exceed B_token.

WIRE: InternalExecutionDagSynthesizer.kt, PartitionClusterManager

SD7-T1-064+mutate-AgentPromptContract [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-021]

R: Research formal Wave Contract prompt formats that block implementation leakage by construction.

I: Inject the clean-room prompt. Block any sibling code. Foreseeable error: provider context windows that drop the interface lattice; place interfaces in high-order immutable slot.

WIRE: AgentPromptContract.kt, DecoupledWorkerSynthesizer

SD7-T1-065+mutate-SpecGraph-throughput [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research streaming atom extraction that removes arbitrary token collection caps for large specifications.

I: Mutate apps/specgraph-foundry compiler `__init__.py` and execution_dag.py to increase throughput and emit partition metrics + boundary hashes. Foreseeable error: memory exhaustion on huge specs; stream rather than materialize.

WIRE: SpecGraphCompiler, InternalExecutionDagSynthesizer

4. 2050 Neuro-Symbolic & Scale-Gap Atoms (Tier-1 Inclusion)

4.1 Inverse Latent Specification Synthesis (ILSS)

SD7-2050-ILSS-01+adversarial-goal-expansion [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-020]

R: Research adversarial counter-specification mining used in security threat modeling and formal methods (STRIDE, attack trees, TLA+ negative properties). Prefer techniques that produce machine-checkable $\neg S$ without requiring a second full LLM call per atom.

I: Before Wave-0, a critic produces Negative Specification $\neg S$ covering Byzantine faults, network partitions, malicious ingress, memory exhaustion, tail-latency SLAs. Every user goal G is paired with anti-goal $\neg G$ that must provably never occur. Foreseeable error: open-ended questions; instead compute Pareto frontier and request single-bit human selection.

WIRE: ContractLatticeSynthesizer, DloiService, FactoryClarificationRequest

SD7-2050-ILSS-02+DLOI-domain-contract-extrapolation [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-2050-ILSS-01]

R: Research automatic grafting of canonical non-functional contracts from knowledge corpora (OWASP, ACID isolation proofs, connection-pool bounds, GC pressure invariants).

I: Query DLOI address map (006 Security, 410 Data Plane, 620 Runtime) and inject mandatory Wave-0 atoms regardless of user prompt brevity. Foreseeable error: over-injection that violates territory; keep injected contracts inside the same partition territory.

WIRE: DloiService, ContractLatticeSynthesizer, SourceAuthorityIndexer

SD7-2050-ILSS-03+epistemic-uncertainty-bounding [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-2050-ILSS-01]

R: Research entropy measures $H(\text{Intent}|\text{Prompt}, \text{DLOI})$ and Pareto-optimal decision matrices for under-determined goals.

I: If entropy $> \epsilon$, compute Pareto frontier (latency vs memory, consistency vs availability, throughput vs complexity) and await single-bit human selection before DAG freeze. Foreseeable error: silent default choices; always surface the matrix.

WIRE: FactoryConfidence, FactoryClarificationRequest, AutonomousWaveOrchestrator

4.2 Formal Concurrency Model Checking

SD7-2050-FMC-01+TLA-Alloy-state-machine [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-020]

R: Research extraction of TLA+ / Alloy specifications from interface contracts and exhaustive checking with TLC or Z3 before any implementation code is written.

I: Every stateful or concurrent atom must emit formal state variables, transition relations, safety and liveness properties. Gate: exhaustive exploration proves zero deadlocks and zero race conditions. Foreseeable error: state-space explosion; bound the model and fail closed if incomplete.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate

SD7-2050-FMC-02+chaos-digital-twin [2050-NEURO-SYMBOLIC]

Track: B

dependsOn: [SD7-T1-060, SD7-T1-061]

R: Research ephemeral containerized chaos runners that inject kernel preemption, socket latency, clock skew, and 95 % memory ceilings while measuring residual memory and error rates.

I: Wave-4 instantiates Digital Twin. Acceptance requires survival of 10 000 chaos transactions with zero thread leaks and residual memory returning to baseline post-GC. Foreseeable error: non-deterministic chaos; seed all PRNGs and record the seed in the evidence bundle.

WIRE: AutonomousWaveOrchestrator, NativeToolchainRegistry, EvidenceCollector

SD7-2050-FMC-03+actor-territory-message-formalism [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-018, SD7-T1-021]

R: Research affine / ownership type systems and disjoint mailbox formalisms that make shared mutable memory across territory boundaries a compile-time error.

I: All inter-actor communication typed as disjoint mailboxes. Violations caught at compile time. Foreseeable error: hidden shared state via statics; ArchitectureComplianceChecker bans static mutable state.

WIRE: ArchitectureComplianceChecker, TerritoryGrant, DecoupledWorkerSynthesizer

4.3 Neuro-Symbolic SMT Verification

SD7-2050-NSV-01+Hoare-logic-annotations [2050-NEURO-SYMBOLIC]

Track: S

dependsOn: [SD7-T1-040]

R: Research machine-checkable pre/post/invariant annotations across Rust (Kani/Creusot), Kotlin, TypeScript, Python (icontract).

I: Every synthesized function must emit requires / ensures / invariant. Functions lacking provable contracts are rejected as unverified. Foreseeable error: annotations that are comments only; require toolchain that discharges them.

WIRE: DecoupledWorkerSynthesizer, LanguageToolchainDriver

SD7-2050-NSV-02+SMT-equivalence-proofs [2050-NEURO-SYMBOLIC]

Track: B

dependsOn: [SD7-2050-NSV-01]

R: Research Z3 / SMT-LIB equivalence proofs on bytecode or AST that discharge arithmetic overflow freedom, array bounds, null dereference impossibility, state-machine validity.

I: $\forall x \in \text{Domain}(\text{requires}): \text{Implementation}(x) \Rightarrow \text{ensures}(x)$. Failure to discharge triggers repair loop before test suite. Foreseeable error: solver timeouts; set strict resource bounds and fail closed.

WIRE: GovernedCompileGate, CompilerDiagnosticRepairLoop**SD7-2050-NSV-03+tri-model-disputation** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** [SD7-T1-040]**R:** Research heterogeneous multi-model synthesis with structural AST edit-distance comparison and adversarial arbitration for high-risk logic.**I:** High-risk state-mutating algorithms synthesized by three independent families. If AST semantics diverge, arbitrator extracts discrepancy and uses SMT oracle to select the superior branch. Foreseeable error: majority vote without proof; always require SMT discharge.**WIRE:** ProviderCascadeRouter, DecoupledWorkerSynthesizer, VerifiedCompletionGate**4.4 Long-Horizon Residual Defect Elimination****SD7-2050-LHD-01+concolic-test-generator** [2050-NEURO-SYMBOLIC]**Track:** B**dependsOn:** [SD7-T1-030, SD7-2050-NSV-02]**R:** Research concolic (concrete + symbolic) test generation that forces exploration of unreached branches via SMT path-condition inversion.**I:** Target 100 % branch and condition coverage (MC/DC for critical lanes). Foreseeable error: path explosion; bound depth and record uncovered residual as Residual Obligation.**WIRE:** Wave1 test generator, VerifiedCompletionGate**SD7-2050-LHD-02+transactional-memory-leak-detector** [2050-NEURO-SYMBOLIC]**Track:** B**dependsOn:** [SD7-2050-FMC-02]**R:** Research byte-accurate heap and descriptor profiling that proves allocations – deallocations = 0 per transaction after warmup.**I:** Any non-zero growth of unreferenced bytes rejects the build as resource-unsound. Foreseeable error: false positives from caches; exclude known warm caches from the calculation.**WIRE:** Digital Twin runner, EvidenceCollector**SD7-2050-LHD-03+residual-obligation-ledger** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** []**R:** Research immutable CAS-chained ledgers of human vetoes that future synthesis engines treat as hard boundaries.**I:** Every human rejection creates a durable formal prohibition clause. Future DAG synthesis reads Residual Obligations as hard constraints. Foreseeable error: loss of vetoes across restarts; store only as CAS-addressed records with exact hashes.**WIRE:** LocalMemoryStore, CAS, RestartCoordinator**SD7-2050-LHD-04+zero-entropy-proof-certificate** [2050-NEURO-SYMBOLIC]**Track:** S**dependsOn:** [SD7-T1-060, SD7-2050-NSV-02, SD7-2050-FMC-01, SD7-2050-FMC-02]**R:** Research machine-checkable cryptographic proof bundles that third parties can validate in seconds without trusting the LLM.**I:** Final deliverable is a certificate vector: SpecHash, ExpandedNegativeSpecHash, SMTVerificationProofsHash, TLAModelCheckReceiptHash, MutationKillScoreHash, DigitalTwinChaosReceiptHash, ToolchainBinaryHash. Foreseeable error: missing fields; VerifiedCompletionGate refuses promotion until all hashes are present.**WIRE:** VerifiedCompletionGate, EvidenceCollector, IntegrityProofs**5. File Constitution & Anti-Boilerplate Atoms (Tier-1)****SD7-T1-070+file-atomicity-constitution** [NEW-SUBSTRATE]**Track:** S**dependsOn:** []**R:** Research single-responsibility file limits and composition-over-inheritance enforcement used in large codebases that survive multi-year autonomous evolution.**I:** Soft 350 LOC / hard 500 LOC. One file = one atomic responsibility. No mixing presentation with decision or execution logic. Enforced by ArchitectureComplianceChecker as deterministic gate. Ban TODO, NotImplemented, fake returns. Foreseeable error: generated files that exceed cap; reject before merge.**WIRE:** ArchitectureComplianceChecker, DecoupledWorkerSynthesizer**SD7-T1-071+AST-stub-hollow-scanner** [NEW-SUBSTRATE]**Track:** S**dependsOn:** [SD7-T1-070]**R:** Research static AST walks (Tree-sitter or language-specific) that reject hollow implementations with zero false negatives on the listed patterns.**I:** Hard rejection of TODO(), NotImplementedError, empty function bodies, empty catch blocks, constant returns, unused parameters in business logic, throw-fake-exception, MockClass in production path. Synthesis rejected immediately, zero reward. Foreseeable error: language-specific AST gaps; start with Kotlin + TypeScript + Python + Rust.**WIRE:** ArchitectureComplianceChecker, DecoupledWorkerSynthesizer**SD7-T1-072+mutation-testing-oracle** [NEW-SUBSTRATE]**Track:** B

dependsOn: [SD7-T1-030]**R:** Research systemic defect injection (negate conditionals, swap operators, invert boundaries, erase calls) and the 85 % kill-rate threshold.**I:** Before atom acceptance inject ≥ 10 mutations. Wave-1 suite must fail on ≥ 85 %. Passing mutated code proves the test is fake and rejects both module and test. Foreseeable error: equivalent mutants; record and quarantine them.**WIRE:** Wave1 test generator, VerifiedCompletionGate

6. Mandatory Caller-Completeness Gate

After all Tier-1 atoms are implemented, execute the following audit before any promotion:

1. Every new class introduced by this DAG (AutonomousWaveOrchestrator, PartitionClusterManager, ContractLatticeSynthesizer, DecoupledWorkerSynthesizer, DependencyInjectionAssembler, NativeToolchainRegistry, all LanguageToolchainDriver implementations, CompilerDiagnosticRepairLoop, and every 2050 atom owner) must appear in at least one production WIRE outside tests.
2. AppSourceTemplate.kt and LanguageScaffold.kt must have zero remaining references in production source.
3. ArchitectureComplianceChecker must report zero violations of the file-atomicity constitution and the AST-stub scanner.
4. A real application (CLI or web) must be synthesized end-to-end through AutonomousWaveOrchestrator and pass VerifiedCompletionGate with a non-empty artifact.
5. The Zero-Entropy Proof Certificate must be present and all hashes non-null.

Failure of any item blocks promotion. This gate is itself an atom:

SD7-T1-099+caller-completeness-gate [NEW-SUBSTRATE]**Track:** S**dependsOn:** [ALL Tier-1 atoms]**R:** Research orphan-detection and caller-graph completeness checks used in large monorepos (Google, Meta, Microsoft) that prevent dead code and unwired new modules.**I:** Implement as an atropos doctor --factory-completeness command that fails CI if any of the five conditions above is false. Foreseeable error: test-only callers counted as production; require callers outside src/test.**WIRE:** CI job, atropos doctor, ArchitectureComplianceChecker

7. Critical Path (Tier-1)

SD7-T1-001 $\rightarrow$ 002 $\rightarrow$ 003 $\rightarrow$ 004SD7-T1-010 $\rightarrow$ 011 $\rightarrow$ (012|013|014|015|016|017) [drivers may proceed in parallel]

SD7-T1-018 (PartitionClusterManager)

SD7-T1-020 $\rightarrow$ 021 $\rightarrow$ 030 $\rightarrow$ 040 $\rightarrow$ 041 $\rightarrow$ 050 $\rightarrow$ 060

SD7-T1-061, 062, 063, 064, 065 may proceed once their direct dependsOn are green.

2050 atoms that depend only on Wave-0 may run in parallel with Wave-2/3 once contracts exist.

SD7-T1-099 is the final gate.

8. Non-Negotiable Mathematical Invariants

 $I(\text{Module_A}; \text{Module_B} \mid \text{Interface_Lattice}) = 0.000 \text{ bits}$ $\text{Territory}(\text{P_r}) \cap \text{Territory}(\text{P_s}) = \emptyset$

Non-zero native exit code = terminal rejection (zero false positives)

 $\text{Worker context} \leq \text{partition footprint}$ (default $\leq 12\,000$ tokens) $C_{\text{real}} = \min(C_{\text{impl}}, C_{\text{int}}, C_{\text{ver}}, C_{\text{ev}})$

False-Green Tolerance = 0.0000 %

Every code byte originates from atomic LLM synthesis against formal interfaces verified by genuine native compilers.

9. Acceptance Criteria for Tier-1 Completion

- AppSourceTemplate.kt and LanguageScaffold.kt are deleted and have zero production references.
- AutonomousWaveOrchestrator can synthesize a non-trivial multi-file application from SpecGraph atoms.
- All generated files pass ArchitectureComplianceChecker (atomicity + stub scanner).
- Real toolchain drivers are invoked; no mock verify.sh remains.
- At least one 2050 neuro-symbolic gate (ILSS or FMC or NSV) is live and blocking.
- Caller-Completeness Gate (SD7-T1-099) is green.
- A Zero-Entropy Proof Certificate is emitted for the synthesized application.

Only after the above is true may Tier-2 atoms be scheduled.

ATROPOS OBLIGATION DAG — FULL MULTI-TIER

Continuation & Completion of Source Document 7 + SD6 Canonical Addenda

Tiers 2 · 3 · 4 + All Deferred Subsystems + Advanced Research Expansions

Maximum atomization · Zero boilerplate · Hybrid Residual RESEARCH+IMPL + HOE Track/WIRE · 2026-10-06

This document is the canonical continuation of the Tier-1 Obligation DAG. It atomizes every deferred domain from Source Document 7 and the SD6 Superiority Addenda (Parts I & II). It is designed to be appended directly beneath Source Document 7.

Extraction rules remain locked: maximum atomization, hybrid format, research prompts force external discovery, impl notes name foreseeable failures and cheapest correct paths that preserve the existing ATROPOS substrate.

TIER-2 — Long-Horizon Governance, Swarm, Storage & Resource Control

Prerequisite: Tier-1 Caller-Completeness Gate (SD7-T1-099) green and a real application has been synthesized end-to-end.

2.1 Strategic Overrides & Architectural Foundations

SD7-T2-001+open-core-airgap-invariant [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research sovereign local-first architectures (Termux, PureOS, GrapheneOS, NixOS offline modes) that keep planning, DAG synthesis and context indexes fully air-gapped while permitting only stateless zero-retention remote inference or CAS storage.

I: Enforce Open-Core Split: core planner, DAG synthesizer, context index and generator remain 100 % open-source local. Hosted plane restricted to optional stateless API or replicated CAS. Air-Gap Invariant: full synthesis and verification operate without network. Foreseeable error: accidental network call inside Wave-0/2; BoundedAgencyGate must refuse any network capability during pure synthesis.

WIRE: BoundedAgencyGate, SourceAuthorityLaw, LocalMemoryStore, CAS

SD7-T2-002+vscode-fork-prohibition [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001]

R: Research headless daemon + thin surface patterns (LSP servers, Language Server Protocol, headless Claude Code / Aider / Codex) that avoid unmaintainable editor forks.

I: Formal rule: do not fork VS Code. Architecture = local headless daemon (atropos.jar) + thin command surfaces (CLI/TUI + local Bridge HTTP/SSE + Compose Desktop + Android APK + optional thin editor window). Agent process lives outside editor process so it survives crashes, runs in CI and executes headless. Foreseeable error: embedding heavy editor process; keep daemon independent.

WIRE: MainActivity (Android), BridgeHttp, CLI entrypoints, atropos.jar packaging

SD7-T2-003+integration-resequencing [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001]

R: Research phased integration ladders used by production agent systems that keep local toolchains as Tier-0 and treat chat/issue trackers as distribution, not core.

I: Resequenced tiers: Tier-0 = Filesystem sandbox + local Git + local toolchains (kotlinc, cargo, tsc, go, gradlew). Tier-1 = Sentry (stack→file:line→territory patch→evidence) + GitHub Actions CI (headless clean-machine reproducibility). Tier-2 = Linear/Jira/Confluence/Datadog/Puppeteer. Tier-3 = Slack/Discord/Community MCP. All integrations under BoundedAgencyGate + SecretSinkMatrix + Territory. Raw tokens never enter prompts. Foreseeable error: elevating chat into production core; refuse.

WIRE: BoundedAgencyGate, SecretSinkMatrix, TerritoryGrant, CI configuration

2.2 Swarm Hierarchy, Coordination & Model Routing

SD7-T2-010+six-level-swarm-topology [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research hierarchical multi-agent topologies (AutoGen, CrewAI, MetaGPT, military command nets) that keep communication $O(N)$ rather than $O(N^2)$ and enforce territory disjointness.

I: Implement 6-level hierarchy: L1 Human Owner/CEO (goal, irreversible approvals, governance mutation); L2 Director Agent (topological decomposition, territory assignment, worktree monitoring, drift detection); L3 Division VP (domain coordination synthesis/verification/storage/delivery); L4 Supervisor Manager (team territory tracking, conflict resolution); L5 Specialist Agent (narrow domain); L6 Worker Agent (isolated atom, zero sibling visibility). Cross-cutting: HR Information Router (audited filtered chokepoint), Auditor (read-only, no execution capture), Custodian (state hygiene, GC, zero LLM overhead). Foreseeable error: flat peer communication; ban and force hierarchical routing.

WIRE: AutonomousWaveOrchestrator, DirectorAgent (new), TerritoryGrant, IsolatedWorktreeService

SD7-T2-011+heterogeneous-provider-matrix [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-010]

R: Research specialized model routing matrices that assign high-velocity drafting, large-context planning, architectural audit, algorithmic cores and zero-cost local validation to distinct providers while surviving 429/offline.

I: Dispatch matrix: Groq Llama-3.3-70B = high-velocity code workers & AST chunk drafting; Gemini 2.5 Pro = large-context spec & DLOI topological planning; Anthropic Claude 3.7 Sonnet = architectural audit, Wave-0 contracts, complex reflexive repair; DeepSeek-V3 = algorithmic routines, heuristic optimization, data structures; Ollama local ARM64 = zero-cost on-device AST validator & security gatekeeper. Fallback: 429 or offline → next provider or local Ollama without pipeline interrupt. Foreseeable error: hard-coding a single provider; RoutePolicy must remain pure data.

WIRE: ProviderCascadeRouter, RoutePolicy, FreeModeGuard

SD7-T2-012+swarm-parallelism-math [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-010]

R: Research Amdahl-derived effective parallelism formulas that subtract coordination overhead, rework conflict and verification debt. Examine real measurements from large agent swarms.

I: $\text{Speedup}(N) = 1 / ((1-p) + p/N)$. $N_{\text{effective}} = N \times (1 - \text{coordinationOverhead}) \times (1 - \text{reworkConflict}) \times (1 - \text{verificationDebt})$. Max parallel lanes = 64 concurrent sub-agents across disjoint worktrees. Critical-path scheduling prioritizes topological bottleneck atoms. $\text{ConflictRisk} \propto \Sigma(|T_i \cap T_j| / |T_i \cup T_j|)$ must remain 0 by territory disjointness. Foreseeable error: optimistic N without measuring overhead; instrument and feed back into $N_{\text{effective}}$.

WIRE: DirectorAgent, PartitionClusterManager, TerritoryGrant

SD7-T2-013+bounded-agency-proposal-gate [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001, SD7-T2-010]

R: Research strongly-typed action proposal pipelines that never execute raw LLM prose or shell commands ambiently (capability systems, object-capability security, sel4-style enforcement).

I: Flow: Model output → untrusted text → deterministic parser → ActionProposal → BoundedAgencyGate → CapabilityEnforcer → TerritoryEnforcer → sandbox execution. No raw LLM prose or shell commands execute ambiently. All actions strongly typed. Foreseeable error: stringly-typed commands; reject any proposal that is not a sealed data class.

WIRE: BoundedAgencyGate, CapabilityEnforcer, TerritoryEnforcer, ActionProposal

2.3 Storage Constitution, Resource Governance & Hardware Health

SD7-T2-020+storage-constitution-laws [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research local-first retention hierarchies and mark-and-sweep GC used by CAS systems (IPFS, Perkeep, Content Addressable Storage in Bazel) that keep phone flash from filling during multi-week autonomous runs.

I: Laws: (01) Global device ceiling $\text{ATROPOS_TOTAL_LOCAL_BYTES} \leq N$ (per-device profile). (02) Retention tiers $\text{HOT} = \text{current checkpoint} + \text{active proposal}$, $\text{WARM} = \text{last } K \text{ successful promotions}$, $\text{COLD} = \text{remote R2 CAS only}$, $\text{EXPIRE} = \text{failed candidates} + \text{rejected proposals} + \text{old worktrees}$. (03) Artifact GC atoms: GC worktrees after promote/reject, GC candidate JARs not in active promotion, cap evidence bundle size (hashes local, bulk on R2). (04) App-factory offload: source+tiny evidence local, full git history and build products to GitHub/R2, local build dirs ephemeral. (05) Pre-flight free-space gate = blocking gate equal to VerifiedCompletionGate. (06) Memory compression: observations become structured bounded records, raw transcripts ephemeral. Foreseeable error: unbounded evidence growth; Custodian must enforce after every wave.

WIRE: Custodian, LocalMemoryStore, CAS, IsolatedWorktreeService, EvidenceCollector

SD7-T2-021+merkle-incremental-build-cache [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T2-020]

R: Research Merkle digest trees for incremental builds (Bazel remote cache, Nix store, content-addressed compilation) that keep rebuild cost $O(\log N)$ at 1 M LOC.

I: Every file has SHA-256. $\text{ModuleHash} = H(\text{Dependencies}) \cup H(\text{Contracts}) \cup H(\text{SourceFiles})$. If ModuleHash unchanged, reuse existing verified binary and skip recompilation. Foreseeable error: hash collisions or partial writes; always verify full content after cache hit.

WIRE: NativeToolchainRegistry, GovernedCompileGate, CAS

SD7-T2-022+thermal-hardware-pacing [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T2-020]

R: Research host-aware throttling for battery, temperature and free RAM used by mobile build systems and long-running Termux agents.

I: Monitor battery level, battery temperature, CPU throttling, free RAM. Throttle law: if $\text{temp} > 42^\circ\text{C}$ or $\text{battery} < 20\%$ reduce parallel workers and pause heavy builds. Ensures safe multi-week execution on pocket hardware. Foreseeable error: ignoring thermal signals; treat as hard gate equal to free-space gate.

WIRE: DirectorAgent, PartitionClusterManager, HostHealthMonitor (new)

SD7-T2-023+mathematical-termination-potential [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research potential functions and oscillation pruning used in formal verification and search algorithms that guarantee termination of autonomous loops.

I: $\Phi(\text{Goal}) = \text{RemainingUnimplementedAtoms} + \text{OpenVerificationGaps} + \text{UnresolvedDiagnostics}$. Transition accepted iff $\Phi(t+1) < \Phi(t)$. If same error seen twice, trigger hypothesis disputation and prune search branch. Prevents infinite autonomous loops by construction. Foreseeable error: non-monotonic progress; log Φ and halt on non-descent.

WIRE: AutonomousWaveOrchestrator, DirectorAgent, VerifiedCompletionGate

SD7-T2-024+quota-economic-governance [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-011]

R: Research inference cost ledgers, free-mode guards and emergency paid gates that prevent quota collapse while preserving local-first preference.

I: Integrate QuotaLedger + FreeModeGuard + EmergencyPaidGate + RoutePolicy. Cascade: local-first → free-ready provider → free-fallback provider → cooldown queue → offline degraded mode → paid emergency only by explicit unlock. 429 causes cooldown not abort. Continuous quality-per-dollar arbitrage. Foreseeable error: silent paid spend; EmergencyPaidGate requires explicit human unlock.

WIRE: ProviderCascadeRouter, RoutePolicy, QuotaLedger (new), FreeModeGuard

SD7-T2-025+exact-hash-restart-continuity [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-020]

R: Research exact-hash restart vectors used by reproducible builds and long-running agent systems that never rebuild goals from model intuition.

I: Restore vector = [RootHumanAuthorityHash, ActiveSourceSpecHash, GoalInvariantFingerprint, DerivationGraphVersion, CurrentAtomId, ActiveTerritory, ActiveContextManifestHash, RemainingBudget, AcceptedFindings, LastVerifiedGate]. Prohibition: never summarize old goal from memory or rebuild from current model intuition. Divergence = blocking halt. Foreseeable error: partial restore; require full vector or refuse.

WIRE: RestartCoordinator, LocalMemoryStore, CAS, SourceAuthorityLaw

TIER-3 — CLI / UNIX Sovereignty + Advanced Anti-Fake-Green Oracles

Prerequisite: Tier-2 storage constitution and swarm topology green.

3.1 Full CLI / UNIX Sovereignty Substrate

SD7-T3-001+pty-process-hosting [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-010]

R: Research isolated POSIX pseudo-terminal (openpty/forkpty) patterns used by tmux, screen, and modern agent CLIs that prevent interactive hangs on npm/cargo/gradle prompts.

I: All external toolchains spawned via isolated PTY. Features: raw termios control, non-blocking I/O poll, max drain timeout 600 000 ms, isolated process group detachment. Foreseeable error: interactive hang; always supply non-interactive flags and kill process group on timeout.

WIRE: BoundedProcessRunner, ToolchainExecutionDriver, all LanguageToolchainDrivers

SD7-T3-002+posix-signal-lifecycle [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-001]

R: Research typed signal handlers that preserve CAS index and Git HEAD integrity under SIGINT/SIGTERM/SIGWINCH/SIGHUP.

I: Handlers: SIGINT → clean worktree restore + record abort evidence; SIGTERM → emit checkpoint CAS snapshot + halt workers; SIGWINCH → dynamic viewport recalculation, no state loss; SIGHUP → background daemon detach. Invariant: signals never corrupt CAS index or Git HEAD. Foreseeable error: unclean shutdown; always write evidence before exit.

WIRE: IsolatedWorktreeService, CAS, RestartCoordinator, Custodian

SD7-T3-003+universal-shell-completion [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research static generation of shell completion from canonical command registries and AST symbol graphs (Click, Cobra, Clap, complete-alias patterns).

I: Generate dynamic flag/argument/file-path/atom autocomplete for Bash, Zsh, Fish, PowerShell directly from CommandRegistry + AstSymbolGraph. No third-party tools. Foreseeable error: stale completions; regenerate on every CommandRegistry change.

WIRE: CommandRegistry, AstSymbolGraph, CLI packaging

SD7-T3-004+cli-tty-discipline [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research strict TTY discipline standards (NO_COLOR, CLICOLOR_FORCE, TERM=dumb, isatty detection) and POSIX exit-code conventions used by professional CLIs.

I: Conformance: NO_COLOR, CLICOLOR_FORCE, TERM=dumb fallback, isatty detection. Output separation: stdout = pure machine JSON/NDJSON or diffs; stderr = telemetry/logs/spinners/diagnostics. POSIX exit codes: 0=verified success, 64=usage/syntax, 65=contract verification failure, 69=unavailable service, 70=internal core fault, 74=IO/storage failure, 77=permission/territory refusal, 78=configuration/policy violation. Foreseeable error: color or progress on stdout; never.

WIRE: CLI entrypoints, all diagnostic emitters

SD7-T3-005+terminal-input-defense [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research bracketed-paste mode and paste-size collapsing used by modern terminals to prevent carriage-return injection and terminal hijacking.

I: Enable ESC[?2004h. Collapse pastes >20 lines into bounded CAS handles. Prevents CR injection and terminal hijacking. Cursor/keyboard: VT100 escapes + UTF-8 raw key decoder + searchable ring buffer. Foreseeable error: accepting multi-line paste as commands; always collapse.

WIRE: CLI input loop, CAS

SD7-T3-006+osc52-clipboard-bridge [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research OSC-52 ANSI clipboard streaming and native fallbacks (wl-copy, xclip, pbcopy, Termux) with secret scrubbing.

I: Protocol: OSC-52 emission. Fallback cascade: Termux clipboard → wl-copy → xclip → pbcopy. Invariant: raw secrets and auth tokens filtered before clipboard write. Foreseeable error: secret leakage; always run SecretSinkMatrix before write.

WIRE: SecretSinkMatrix, CLI surfaces

SD7-T3-007+standalone-virtual-pager [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research zero-dependency terminal pagers with half-page scroll, regex search and unified-diff syntax highlighting.

I: Internal zero-dependency pager: forward/backward half-page scroll, regex search, unified-diff syntax highlighting, non-blocking escape. Foreseeable error: external less/more dependency; keep pure internal.

WIRE: CLI surfaces, diagnostic viewers

3.2 Advanced Anti-Fake-Green & Anti-Boilerplate Oracles

SD7-T3-010+property-based-fuzzing-oracle [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-030, SD7-T1-072]

R: Research property-based fuzzing across extreme integers, Unicode stress strings, recursive JSON structures, null-pointer injections and race-condition concurrency (Hypothesis, QuickCheck, cargo-fuzz patterns).

I: Automated fuzz test cases across type domains. Predicate: module must satisfy formal invariants across 1 000 deterministic PRNG iterations. Foreseeable error: non-deterministic seeds; record seed in evidence bundle.

WIRE: Wave-1 test generator, VerifiedCompletionGate, EvidenceCollector

SD7-T3-011+callgraph-reachability-proof [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-071]

R: Research directed AST call-graph traversal that proves every synthesized symbol has an unbroken path from application entry point (main()) or public RPC/HTTP router).

I: Every synthesized class/function/port must have unbroken path from entry point. Isolated orphan symbols without invocation edges trigger immediate build rejection. Zero dead-code tolerance. Foreseeable error: reflection/dynamic dispatch hiding edges; require explicit registration or fail.

WIRE: ArchitectureComplianceChecker, VerifiedCompletionGate, AstSymbolGraph

SD7-T3-012+dual-oracle-differential-compilation [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-010, SD7-T1-061]

R: Research dual-toolchain differential compilation (kotlinc JVM vs GraalVM Native, tsc vs OXC/SWC, go vs go vet -race) that eliminates runtime-specific false greens.

I: Compile and execute suite under two distinct toolchains/modes. Divergent behavior or compiler-specific green fails closed. Foreseeable error: accepting single-toolchain success; always require both oracles.

WIRE: NativeToolchainRegistry, VerifiedCompletionGate, GovernedCompileGate

TIER-4 — Cloud Delivery, Schema Compilation & Autonomous Deployment

Prerequisite: Tier-3 CLI sovereignty and anti-fake-green oracles green.

4.1 Storage Schema Compilation

SD7-T4-001+StorageSchemaCompiler [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-020]

R: Research generation of relational/NoSQL DDL, indexes and typed ORM/client mappings that stay bit-identical to Wave-0 contracts (SQLAlchemy, Prisma, Room, Exposed, SQLx patterns).

I: Generate relational/NoSQL DDL, database indexes, typed ORM/client query mappings matching Wave-0 contracts. ATOM-L1-003. Foreseeable error: schema drift from contracts; regenerate from contracts only, never hand-edit.

WIRE: ContractLatticeSynthesizer, SupabaseSchemaDriver, VerifiedCompletionGate

4.2 Cloud Deployment Drivers

SD7-T4-010+CloudDeploymentAdapter [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research uniform non-mock delivery seams that abstract Vercel/Netlify/Supabase/local-dist behind a single contract.

I: Uniform contract for non-mock deployment to hosting providers. No shell scripts that echo success. Foreseeable error: mock deploy that returns fake URL; require real HTTP 200 health-check.

WIRE: AutonomousWaveOrchestrator, all concrete drivers

SD7-T4-011+VercelDeploymentDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-010]

R: Research direct REST SHA-1 content-addressed upload to Vercel Deployments API with manifest synthesis and production URL extraction.

I: Direct API deployer packaging build outputs to Vercel production endpoints. SHA-1 content-addressed streaming. Extract live production URL. Foreseeable error: accepting non-200; fail closed until health-check passes.

WIRE: CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-012+NetlifyDeploymentDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-010]

R: Research ZIP REST deployment to Netlify API with production URL extraction.

I: Direct ZIP REST deployer. Same health-check invariant. Foreseeable error: missing site ID or token; fail closed with clear diagnostic.

WIRE: CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-013+SupabaseSchemaDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-001, SD7-T4-010]

R: Research automated PostgreSQL migration runners with Row-Level Security policy synthesis and connection-pool bounds.

I: Applies SQL migrations, generates RLS policies, registers DB configurations. Enforce connection-pool bounds from Wave-0 contracts. Foreseeable error: open RLS; default deny and require explicit policies.

WIRE: StorageSchemaCompiler, CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-014+AutonomousDeliveryEngine [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T4-010, SD7-T4-011, SD7-T4-012, SD7-T4-013]

R: Research autonomous deployment pipelines that assert HTTP 200 health checks at production domains or bit-verified binary on disk before declaring success.

I: ATOM-L4-003. Autonomous deployment pipeline. Acceptance = HTTP 200 live health-check at target production URL or bit-verified ELF/APK on disk.

Foreseeable error: declaring success on upload alone; always require health-check or binary proof.

WIRE: AutonomousWaveOrchestrator, CloudDeploymentAdapter, VerifiedCompletionGate, EvidenceCollector

SPECGRAPH FOUNDRY — Expanded Ingestion Atoms

These expand the over-condensed SD7-T1-065 into full maximum atomization.

SD7-SG-001+unbounded-ingestion-pipeline [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research multi-gigabyte streaming parsers that avoid stack recursion crashes (iterative tokenization, chunked SHACL validation, memory-mapped reading).

I: Concrete mutations to statement_segmentation.py and atomic_decomposition.py for multi-gigabyte spec parsing without stack recursion. Stream rather than materialize. Foreseeable error: recursion depth; convert all recursive descent to iterative.

WIRE: SpecGraphCompiler, atomic_decomposition.py, statement_segmentation.py

SD7-SG-002+SHACL-validation-proof-bundles [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-SG-001]

R: Research RDF/SHACL shape conformance and feedback-arc-set reduction for formal dependency graphs.

I: Mutations to dependency_compiler.py and shacl_validation.py for RDF/SHACL shape conformance and feedback arc set reduction. Emit formal proof bundles. Foreseeable error: cycles surviving; fail closed on any remaining arc.

WIRE: SpecGraphCompiler, dependency_compiler.py, shacl_validation.py, InternalExecutionDagSynthesizer

ADVANCED RESEARCH EXPANSIONS (Beyond Source Doc 7)

These atoms advance the architecture with 2026-era techniques discovered by external research pressure.

SD7-ADV-001+capability-based-security-model [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-013]

R: Research object-capability (ocap) and seL4-style capability systems that can be applied to LLM action proposals so that possession of a capability is the only authority.

I: Elevate BoundedAgencyGate to a pure capability system. Every ActionProposal must carry the exact set of capabilities required; missing capability = hard reject.

Foreseeable error: ambient authority; purge any ambient capability.

WIRE: BoundedAgencyGate, CapabilityEnforcer, ActionProposal

SD7-ADV-002+content-addressed-evidence-merkle [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-020, SD7-T2-021]

R: Research pure content-addressed evidence ledgers that make every verification receipt independently verifiable by third parties without trusting ATROPOS.

I: Every evidence bundle is a Merkle tree of hashes. Third-party verifier can validate the Zero-Entropy Proof Certificate in seconds. Foreseeable error: mutable evidence; all evidence must be append-only CAS.

WIRE: EvidenceCollector, CAS, IntegrityProofs, VerifiedCompletionGate

SD7-ADV-003+formal-wave-invariant-checker [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-060, SD7-2050-FMC-01]

R: Research static checkers that prove each wave respects the clean-room mutual-information invariant and territory disjointness before any LLM call is issued.

I: Before Wave-2 dispatch, a formal checker proves $I(f_j; f_k|I)=0$ and Territory disjointness. Failure aborts the wave. Foreseeable error: runtime-only checks; move to pre-dispatch static proof.

WIRE: AutonomousWaveOrchestrator, PartitionClusterManager, ArchitectureComplianceChecker

SD7-ADV-004+provider-quality-arbitrage-ledger [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-024]

R: Research continuous quality-per-dollar measurement and Pareto routing across heterogeneous providers that updates after every verified atom.

I: After every successful VerifiedCompletionGate, record (provider, token cost, mutation-kill score, SMT discharge time). RoutePolicy continuously re-ranks. Foreseeable error: stale rankings; decay half-life of 48 h.

WIRE: RoutePolicy, QuotaLedger, ProviderCascadeRouter, EvidenceCollector

SD7-ADV-005+deterministic-replay-harness [AMENDMENT-CANDIDATE]

Track: B

dependsOn: [SD7-T2-025, SD7-T3-010]

R: Research deterministic replay of entire multi-week autonomous runs from exact-hash vectors and recorded PRNG seeds so that any failure can be reproduced bit-identically.

I: Every run records the full restore vector + all PRNG seeds + provider response hashes. Replay harness can re-execute any partition offline. Foreseeable error: non-deterministic external services; pin all responses via CAS.

WIRE: RestartCoordinator, EvidenceCollector, CAS, IsolatedWorktreeService

CROSS-TIER ACCEPTANCE GATES

SD7-GATE-T2+tier2-completion [NEW-SUBSTRATE]

| Maximum atomization | Hybrid Residual+HOE | All tiers complete

Track: S

dependsOn: [ALL Tier-2 atoms]

R: Research completeness criteria for long-horizon governance layers that must be live before CLI sovereignty is activated.

I: Tier-2 complete when: (1) Open-Core + Air-Gap invariants enforced, (2) 6-level swarm topology live with territory disjointness, (3) Storage Constitution Laws 01–06 enforced by Custodian, (4) Thermal pacing and Φ potential active, (5) Exact-hash restart continuity proven by forced restart test, (6) QuotaLedger prevents silent paid spend. Foreseeable error: partial governance; all six must be green.

WIRE: CI job, atropos doctor --tier2-completeness

SD7-GATE-T3+tier3-completion [NEW-SUBSTRATE]

Track: S

dependsOn: [ALL Tier-3 atoms, SD7-GATE-T2]

R: Research completeness criteria for sovereign terminal + anti-fake-green layers.

I: Tier-3 complete when: (1) All external toolchains run under PTY with signal lifecycle, (2) Shell completion generated for four shells, (3) TTY discipline and exit codes enforced, (4) Property-based fuzzing, call-graph reachability and dual-oracle differential all blocking, (5) Zero orphan symbols in a synthesized application. Foreseeable error: accepting single-toolchain success; dual oracle mandatory.

WIRE: CI job, atropos doctor --tier3-completeness

SD7-GATE-T4+tier4-completion [NEW-SUBSTRATE]

Track: S

dependsOn: [ALL Tier-4 atoms, SD7-GATE-T3]

R: Research end-to-end delivery acceptance that requires live production health-check or bit-verified binary.

I: Tier-4 complete when: (1) StorageSchemaCompiler emits DDL matching Wave-0, (2) At least one cloud driver (Vercel or Netlify) deploys a real application and receives HTTP 200, (3) SupabaseSchemaDriver applies RLS policies, (4) AutonomousDeliveryEngine emits Zero-Entropy Proof Certificate containing deployment URL or binary hash. Foreseeable error: upload-only success; health-check mandatory.

WIRE: CI job, atropos doctor --tier4-completeness, AutonomousDeliveryEngine

CRITICAL PATH ACROSS TIERS

Tier-1 (already delivered) → SD7-GATE-T1 (Caller-Completeness) →

Tier-2 governance + swarm + storage → SD7-GATE-T2 →

Tier-3 CLI sovereignty + advanced oracles → SD7-GATE-T3 →

Tier-4 cloud delivery + schema → SD7-GATE-T4 →

Full system accepted only when all four gates are green and a multi-week autonomous run has produced a Zero-Entropy Proof Certificate.

NON-NEGOTIABLE INVARIANTS (ALL TIERS)

I(Module_A; Module_B | Interface_Lattice) = 0.000 bits

Territory(P_r) ∩ Territory(P_s) = ∅

Non-zero native exit code = terminal rejection

False-Green Tolerance = 0.0000 %

Air-Gap Invariant for pure synthesis and verification

Exact-hash restart continuity; never rebuild goal from model intuition

Every code byte originates from atomic LLM synthesis against formal interfaces verified by genuine native compilers

C_real = min(C_impl, C_int, C_ver, C_ev)

$\Phi(t+1) < \Phi(t)$ monotonic descent or halt

END OF FULL MULTI-TIER OBLIGATION DAG

Designed for direct append beneath Source Document 7 · Maximum atomization of every deferred domain · Advanced research expansions included · 2026-10-06

ATROPOS RESIDUAL OBLIGATION DAG

Final Missed-Atom Inventory — Byte / Token / Parameter / File Level

Continues directly from Tier-1 Hybrid DAG + Full Multi-Tier DAG

Maximum atomization of every omitted item · Outside-the-box mathematical & systems advances · 2026-10-06

This document is the third and final Obligation DAG in the Source Document 7 series. It atomizes every residual item identified by byte-level audit against SD7, SD6 Addenda Parts I & II, the 2050 Scale-Gap Assessment, and the concrete Kotlin/Python reference implementations. It leaves off exactly where the previous two DAGs ended.

Extraction rules locked: maximum atomization, hybrid Residual RESEARCH+IMPL + HOE Track/dependsOn/WIRE, research forces external discovery, impl notes name foreseeable failures and cheapest correct paths.

1. Runtime Byte Ceilings & Throttle Removals (SD7 Diagnostic Core)

SD7-R-001+lift-contextCapBytes [RESIDUAL]

Track: S

dependsOn: []

R: Research dynamic context-window governance used by long-context agent runtimes (Claude Code, Aider, Cursor, Continue.dev) that raise or remove hard byte ceilings without risking OOM on constrained hosts.

I: Mutate AgentContextCollector.kt: replace hardcoded contextCapBytes (4 KB–64 KB) with a profile-driven configurable ceiling that defaults to partition footprint ($\leq 12\,000$ tokens) and can be lifted for factory runs. Foreseeable error: unbounded growth on Termux; keep a hard device-profile ceiling and refuse synthesis if free RAM is below threshold.

WIRE: AgentContextCollector.kt, HostHealthMonitor, PartitionClusterManager

SD7-R-002+lift-maxAdvances [RESIDUAL]

Track: S

dependsOn: [SD7-R-001]

R: Research loop-limit designs that prevent infinite agent cycles while still permitting multi-thousand-step autonomous factory runs.

I: Mutate SelfHostRuntimeRunLimits.kt: replace hardcoded maxAdvances = 15 with a wave-aware limit derived from remaining Φ potential and partition count. Factory mode must allow $\geq 10\,000$ advances. Foreseeable error: silent infinite loop; couple to $\Phi(t+1) < \Phi(t)$ and oscillation prune.

WIRE: SelfHostRuntimeRunLimits.kt, AutonomousWaveOrchestrator, DirectorAgent

SD7-R-003+lift-GlobalByteCeiling [RESIDUAL]

Track: S

dependsOn: [SD7-R-001]

R: Research global memory/context ceilings that scale with available host resources while preserving the clean-room token bound $O(|I|+|a_i|)$.

I: Introduce a GlobalByteCeiling policy atom that is raised for factory mode to permit 50 k–100 k+ LOC ASTs. Still enforce per-worker context convexity. Foreseeable error: single worker saturating host RAM; keep per-partition bound absolute.

WIRE: ExecutionPolicyEngine, AgentContextCollector, HostHealthMonitor

SD7-R-004+deprecate-atoms-md-dump [RESIDUAL]

Track: S

dependsOn: [SD7-T1-003]

R: Research the separation of audit logging from synthesis authority so that atoms written to .atropos/research/atoms.md never become the source of truth for file generation.

I: Refactor or deprecate the path that dumps SpecGraph atoms into .atropos/research/atoms.md while templates still emit files. After Tier-1 template purge, the dump becomes pure audit; synthesis authority is the Wave orchestrator. Foreseeable error: dual sources of truth; ArchitectureComplianceChecker must reject any generation path that reads the dump.

WIRE: AppFactoryRouter, SpecGraphCompiler, ArchitectureComplianceChecker

2. Missing Structural Files & Explicit Code Subsystems

SD7-R-010+ToolchainExecutionDriver [RESIDUAL]

Track: S

dependsOn: [SD7-T1-010]

R: Research sandboxed process harnesses that capture exact stdout/stderr/binary artifacts under timeout, byte and line limits (Bazel sandbox, Bubblewrap, Firejail patterns).

I: Create src/main/kotlin/atropos/core/toolchain/ToolchainExecutionDriver.kt as the dedicated execution harness. Responsibilities: process sandboxing, raw stdout/stderr capture, timeouts, byte limits, clean env. ATOM-SYS-TC-001. Foreseeable error: leaking host env; always start with empty environment and inject only required vars.

WIRE: NativeToolchainRegistry, BoundedProcessRunner, all LanguageToolchainDrivers

SD7-R-011+BehavioralTestSynthesizer [RESIDUAL]

Track: S

dependsOn: [SD7-T1-020, SD7-T1-030]

R: Research dedicated mock-and-invariant test generators that produce suites guaranteed to fail against empty stubs and compile against Wave-0 contracts only.

I: Create src/main/kotlin/atropos/core/factory/BehavioralTestSynthesizer.kt (ATOM-L2-001). Responsibility: mock synthesis + invariant test generation for Wave-1. Must not be collapsed into a generic wave generator. Foreseeable error: tests that pass vacuously; require at least one failing assertion on a stub.

WIRE: ContractLatticeSynthesizer, VerifiedCompletionGate, Wave-1 pipeline

SD7-R-012+mutate-AppProjectSpecParser [RESIDUAL]

Track: S

dependsOn: [SD7-T1-003]

R: Research intent parsers that never short-circuit atom graphs into high-level AppIntent objects that bypass SpecGraph.

I: Mutate or replace AppProjectSpecParser.kt so that parsed intent always flows into SpecGraph atomization; never into a template path. Foreseeable error: residual AppIntent-only branch; delete it.

WIRE: AppFactoryRouter, SpecGraphCompiler, AutonomousWaveOrchestrator

SD7-R-013+DeterministicChecks [RESIDUAL]

Track: S

dependsOn: [SD7-T1-061]

R: Research deterministic AST symbol resolution and package-to-directory alignment checkers used by production monorepos.

I: Ensure DeterministicChecks.kt is wired as a mandatory gate inside VerifiedCompletionGate: AST symbols resolve, packages match directory structures. Foreseeable error: optional check; make it blocking.

WIRE: VerifiedCompletionGate, ArchitectureComplianceChecker, AstSymbolGraph

SD7-R-014+semantic-types-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research orthogonal semantic-type taxonomies used by formal requirement engines that prevent dimension collapse.

I: Mutate apps/specgraph-foundry/.../semantic_types.py so that every atom carries a stable semantic type used by Wave-0 dimension filters (INTERFACE, PROTOCOL, SCHEMA, DATA_MODEL). Foreseeable error: free-text dimensions; constrain to closed enum.

WIRE: SpecGraphCompiler, ContractLatticeSynthesizer

SD7-R-015+requirement-quality-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research lexical vagueness detectors (VAGUE_TERMS) and requirement quality scoring used by aerospace/medical requirements tools.

I: Mutate requirement_quality.py to flag VAGUE_TERMS and reject or escalate under-specified atoms before Wave-0. Foreseeable error: silent acceptance of vague goals; route to ILSS epistemic bounding.

WIRE: SpecGraphCompiler, ILSS pipeline, FactoryClarificationRequest

SD7-R-016+unresolved-tracker-py [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research ungrounded-requirement trackers that keep a durable ledger of atoms lacking provenance.

I: Mutate unresolved_tracker.py so that every ungrounded candidate is recorded and blocks Phase-100 until resolved or explicitly waived. Foreseeable error: lost unresolved set across restarts; store in CAS.

WIRE: SpecGraphCompiler, ResidualObligationLedger, CAS

SD7-R-017+Sd5HumanIntentInvariants-ILSS-bridge [RESIDUAL]

Track: S

dependsOn: [SD7-2050-ILSS-01, SD7-2050-ILSS-03]

R: Research authority lattices that keep human prompt primacy (HIA-001–HIA-008) while still permitting adversarial goal expansion.

I: Bridge Sd5HumanIntentInvariants.kt into the ILSS pipeline so that rank(research) < rank(human_intent) is enforced even when ¬S is generated. Foreseeable error: ILSS silently redefining objectives; SourceAuthorityLaw must reject.

WIRE: Sd5HumanIntentInvariants.kt, ILSS, SourceAuthorityLaw, FactoryClarificationRequest

SD7-R-018+TreeSitterGrammarBridge [RESIDUAL]

Track: S

dependsOn: [SD7-T1-071]

R: Research production Tree-sitter grammar bridges that expose a uniform AST walk API across Kotlin, TypeScript, Python, Rust, Go.

I: Create the concrete TreeSitterGrammarBridge class mandated by SD6 EXTIRPATION_01. All AST stub/hollow scans and call-graph walks must go through this bridge. Foreseeable error: language-specific ad-hoc parsers; single bridge only.

WIRE: ArchitectureComplianceChecker, call-graph reachability, AST-stub scanner

3. Low-Level OS Sandboxing & Process Jailing

SD7-R-020+chroot-unshare-jail [RESIDUAL]

Track: B

dependsOn: [SD7-R-010, SD7-T3-001]

R: Research Linux namespace isolation (unshare, chroot, bubblewrap, Firejail, landlock) that provides READ_ONLY_ROOT + WRITEABLE_WORKTREE with minimal environment.

I: Execute every compiler/toolchain inside chroot/unshare or container sandbox. Filesystem policy: READ_ONLY_ROOT_DIRECTORY (/) + only the isolated worktree directory mounted writeable. Secrets stripped. Foreseeable error: host filesystem escape; fail closed on any path outside the worktree.

WIRE: ToolchainExecutionDriver, BoundedProcessRunner, IsolatedWorktreeService

4. Delivery Platforms — CONTAINER & LOCAL_DIST

SD7-R-030+TargetPlatform-CONTAINER [RESIDUAL]

Track: B

dependsOn: [SD7-T4-010]

R: Research deterministic OCI/container image assembly from verified build artifacts (Buildah, kaniko, Docker BuildKit rootless).

I: Add TargetPlatform.CONTAINER. Produce bit-verified OCI image with SHA-256 manifest. Foreseeable error: non-reproducible layers; pin base images by digest.

WIRE: CloudDeploymentAdapter, AutonomousDeliveryEngine, TargetPlatform enum

SD7-R-031+TargetPlatform-LOCAL_DIST [RESIDUAL]

Track: B

dependsOn: [SD7-T4-010]

R: Research local distribution packaging that emits ELF/APK/JAR + SHA-256 proof manifests to a dist/ directory.

I: Add TargetPlatform.LOCAL_DIST (or LOCAL_BINARY). Package to disk with bit-level SHA-256 proof manifests. Acceptance = valid signed/bit-verified binary on disk. Foreseeable error: missing proof file; VerifiedCompletionGate refuses.

WIRE: CloudDeploymentAdapter, AutonomousDeliveryEngine, TargetPlatform enum, IntegrityProofs

SD7-R-032+Google-Drive-CAS-cold-tier [RESIDUAL]

Track: B

dependsOn: [SD7-T2-020]

R: Research Google Drive as an additional content-addressed cold-tier backend alongside R2.

I: Storage Constitution COLD tier: REMOTE_R2_OR_DRIVE_CAS_BLOBS. Implement Drive CAS adapter for cold evidence and build products. Foreseeable error: Drive API rate limits; treat as best-effort with R2 primary.

WIRE: CAS, Custodian, Storage Constitution

SD7-R-033+COMPILER_TEMP_JARS-eviction [RESIDUAL]**Track:** B**dependsOn:** [SD7-T2-020]**R:** Research automatic eviction of compiler temporary JARs and intermediate build products.**I:** EXPIRE retention tier must explicitly include COMPILER_TEMP_JARS. Custodian sweeps them after every wave. Foreseeable error: leftover temp JARs filling flash; include in mark-and-sweep.**WIRE:** Custodian, Storage Constitution, IsolatedWorktreeService

5. Schema-Driven Type Serialization Integrity (INVARIANT_04)

SD7-R-040+runtime-codec-synthesis [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-020, SD7-T4-001]**R:** Research mandatory round-trip serializer generation (Zod, Pydantic, Serde, kotlinx.serialization) that enforces ingress/egress validation, unknown-field rejection and fail-closed overflow.**I:** Wave-0 or Wave-3 must synthesize bidirectional runtime codecs for every boundary type. Rules: all ingress/egress validated at runtime boundary; unknown fields rejected; payload overflow fails closed at parser entry. Foreseeable error: optional codecs; make them mandatory for any public boundary.**WIRE:** ContractLatticeSynthesizer, DependencyInjectionAssembler, ArchitectureComplianceChecker

6. Mathematical & Information-Theoretic Operators (AACS-2036 Part 0)

SD7-R-050+memory-fold-operator [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-040, SD7-T1-018]**R:** Research internal state-transition operators for isolated agents that fold successive atoms without reloading the full codebase (memory-fold, recurrent state machines, linear attention folds).**I:** Implement $\sigma_{\{t+1\}} = \delta(\sigma_t, a_{\{t\}}, I)$ inside each partition worker so that topological atom sequence is processed with constant context. Foreseeable error: context growth; enforce that $|\sigma|$ stays bounded by partition footprint.**WIRE:** DecoupledWorkerSynthesizer, PartitionClusterManager**SD7-R-051+context-convexity-bound** [RESIDUAL]**Track:** S**dependsOn:** [SD7-R-050]**R:** Research formal proofs that context remains $O(|I| + |a_i|)$ under ancestral interface inclusion.**I:** Enforce $X(\tau_i) = I \cup \{a_i\} \cup \text{Ancestors}(a_i, I)$. Any prompt that exceeds this bound is rejected before provider dispatch. Foreseeable error: accidental inclusion of sibling code; scrub and assert.**WIRE:** AgentPromptContract, DecoupledWorkerSynthesizer, ArchitectureComplianceChecker**SD7-R-052+zero-entropy-constraint** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-060]**R:** Research operational definitions of $\text{Entropy}(C \mid S, I) = 0$ that can be approximated by the conjunction of all verification oracles.**I:** The synthesis operator Φ must satisfy $\text{Entropy}(C \mid S, I) = 0$ as the formal objective. Approximated by the product of all verifiers = 1 and the Zero-Entropy Proof Certificate. Foreseeable error: treating it as rhetoric; wire it as a blocking predicate in Phase-100.**WIRE:** VerifiedCompletionGate, Zero-Entropy Proof Certificate**SD7-R-053+edge-qualified-decoupling** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-021]**R:** Research mutual-information nullification conditioned on the absence of a dependency edge.**I:** $I(f_j; f_k \mid I) = 0 \ \forall j \neq k$ such that $(v_j, v_k) \notin E_A$. The edge condition must be checked by the partitioner; workers that share an edge may see ancestral contracts only. Foreseeable error: ignoring the edge qualifier; enforce it in Ψ .**WIRE:** PartitionClusterManager, AgentPromptContract**SD7-R-054+verifier-ensemble-product** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-061]**R:** Research product calculi over deterministic verifier ensembles that refuse partial success.**I:** $\prod_{v \in V} v(\Phi(P_r, I)) = 1$ where $V = \{v_{\text{syntax}}, v_{\text{type}}, v_{\text{property}}, v_{\text{boundary}}\}$. Any single 0 fails the partition. Foreseeable error: soft-AND; implement hard product.**WIRE:** VerifiedCompletionGate, GovernedCompileGate**SD7-R-055+error-gradient-diff-subspace** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T1-041]**R:** Research repair loops that restrict mutations to the territory of the failing partition (diff-subspace restriction).**I:** $\nabla E = \text{Diagnostics}(v)$ and $\Delta f \subseteq \text{Territory}(P_r)$. Repair must never touch files outside the partition territory. Foreseeable error: global search-and-replace; enforce territory lock.**WIRE:** CompilerDiagnosticRepairLoop, TerritoryGrant, IsolatedWorktreeService**SD7-R-056+swarm-acceptance-volume** [RESIDUAL]**Track:** S**dependsOn:** [SD7-T2-012]**R:** Research four-variable acceptance volume operators used by large agent systems.**I:** $V_{\text{accepted}} = \min(V_{\text{generated}}, V_{\text{integrated}}, V_{\text{verified}}, V_{\text{evidenced}})$. Wire into Director metrics. Foreseeable error: counting generated lines as accepted; always take the min.**WIRE:** DirectorAgent, EvidenceCollector, VerifiedCompletionGate**SD7-R-057+phase100-acceptance-velocity** [RESIDUAL]**Track:** S**dependsOn:** [SD7-R-056, SD7-T2-023]

R: Research Phase-100 predicates and acceptance-velocity metrics that declare a multi-week run complete.

I: Phase 100 = $\wedge(\text{MandatoryPredicates}) \wedge \text{RuntimeAcceptance} \wedge \text{NegativePathTests} \wedge \text{EvidenceComplete} \wedge \neg \text{FakeSuccess} \wedge \neg \text{CriticalOrphan}$. $\text{AcceptanceVelocity} = \Delta \text{VerifiedAcceptancePredicates} / \Delta \text{Days}$. Both must be computed and stored in the governance ledger. Foreseeable error: declaring success on compile alone; require full conjunction.

WIRE: VerifiedCompletionGate, DirectorAgent, GovernanceLedger, EvidenceCollector

SD7-R-058+partition-scaling-complexity [RESIDUAL]

Track: S

dependsOn: [SD7-T1-018]

R: Research complexity proofs that synthesized partitions scale $O(N/K)$ independent of total system lines.

I: Document and assert that partition work is $O(N/K)$. Instrument and fail CI if measured complexity exceeds the bound by a constant factor. Foreseeable error: quadratic hidden costs; measure end-to-end.

WIRE: PartitionClusterManager, DirectorAgent

7. Concrete Kotlin Data Contracts, Enums & HTTP Wire Specs

SD7-R-060+ExtractionAtom-model [RESIDUAL]

Track: S

dependsOn: []

R: Research minimal complete atom models that carry id, statement, dimension, dependencies, estimatedTokens, targetPath.

I: Define data class ExtractionAtom(id, statement, dimension, dependencies, estimatedTokens, targetPath) as the canonical atom type used by every downstream component. Foreseeable error: ad-hoc maps; single sealed model only.

WIRE: SpecGraphCompiler, PartitionClusterManager, DecoupledWorkerSynthesizer

SD7-R-061+SynthesisPartition-model [RESIDUAL]

Track: S

dependsOn: [SD7-R-060]

R: Research partition models that carry partitionId, atoms, targetFiles, requiredContracts, dependencies, totalEstimatedTokens.

I: Define data class SynthesisPartition(...) exactly. All Ψ output must be this type. Foreseeable error: lossy intermediate representations; preserve full fidelity.

WIRE: PartitionClusterManager, DecoupledWorkerSynthesizer, AutonomousWaveOrchestrator

SD7-R-062+RepairLoopResult-ToolchainResult [RESIDUAL]

Track: S

dependsOn: [SD7-T1-041]

R: Research structured result types that carry success, attempts, diagnostics, exitCode, and full diagnostic maps (file, line, column, error_code, trace).

I: Define RepairLoopResult(success, attempts, diagnostics) and ToolchainResult(passed, exitCode, diagnostics). Diagnostic map must include error_code and trace. Return status CONVERGED on success. Foreseeable error: stringly diagnostics; structured only.

WIRE: CompilerDiagnosticRepairLoop, ToolchainExecutionDriver, VerifiedCompletionGate

SD7-R-063+TargetPlatform-enum-complete [RESIDUAL]

Track: S

dependsOn: [SD7-R-030, SD7-R-031]

R: Research exhaustive target-platform enums that include LOCAL_BINARY, ANDROID_APK, VERCEL, NETLIFY, CONTAINER, and any future LOCAL_DIST alias.

I: enum class TargetPlatform { LOCAL_BINARY, ANDROID_APK, VERCEL, NETLIFY, CONTAINER }. All delivery drivers must pattern-match exhaustively. Foreseeable error: silent fall-through; use sealed when.

WIRE: AutonomousWaveOrchestrator, CloudDeploymentAdapter, AutonomousDeliveryEngine

SD7-R-064+BuildExecutionReport-sealed [RESIDUAL]

Track: S

dependsOn: [SD7-T1-060]

R: Research sealed report hierarchies with explicit typed phase failure constants.

I: BuildExecutionReport = Success(totalFilesGenerated, artifactOnDisk, deploymentUrl) | Halted(phase, diagnostics). Phase strings must be the closed set: WAVE_0_FAILURE, WAVE_3_SYNTHESIS_FAILURE, WAVE_4_ASSEMBLY_FAILURE, GLOBAL_COMPILE_FAILED, GLOBAL_TEST_FAILED, PACKAGING_FAILED. Foreseeable error: free-text phases; constrain.

WIRE: AutonomousWaveOrchestrator, VerifiedCompletionGate

SD7-R-065+Wave0-dimension-filter [RESIDUAL]

Track: S

dependsOn: [SD7-R-014, SD7-T1-020]

R: Research dimension filters that select only INTERFACE/PROTOCOL/SCHEMA/DATA_MODEL atoms for Wave-0.

I: Wave-0 contracts = atoms.filter { it.dimension in setOf("INTERFACE", "PROTOCOL", "SCHEMA", "DATA_MODEL") }. Enforce closed set. Foreseeable error: leaking implementation atoms into Wave-0; filter strictly.

WIRE: ContractLatticeSynthesizer, SpecGraphCompiler

SD7-R-066+directed-RepairNode-on-edge [RESIDUAL]

Track: S

dependsOn: [SD7-T1-060, SD7-R-055]

R: Research directed repair nodes that target a single failing integration edge instead of aborting the entire build.

I: If an integration edge breaks in Wave-4, create a directed RepairNode in the DAG targeted specifically at the failing edge; do not abort the whole build. Foreseeable error: global abort; schedule edge-local repair.

WIRE: AutonomousWaveOrchestrator, CompilerDiagnosticRepairLoop, DirectorAgent

SD7-R-067+Vercel-REST-wire-contract [RESIDUAL]

Track: B

dependsOn: [SD7-T4-011]

R: Research the exact Vercel Deployments API contract used by production deployers.

I: File upload: POST https://api.vercel.com/v2/files?teamId=... with header x-vercel-digest: <sha1>. Deployment manifest: POST https://api.vercel.com/v13/deployments?teamId=... with projectSettings.framework = null. HTTP client timeout Duration.ofSeconds(30). Foreseeable error: wrong endpoint version; pin the documented paths.

WIRE: VercelDeploymentDriver, SecretSinkMatrix

8. Hardcoded Numerical Limits, Timeouts & Benchmark Criteria

SD7-R-070+AndroidSdkDriver-concrete-thresholds [RESIDUAL]

Track: B

dependsOn: [SD7-T1-017]

R: Research production Android build timeouts and buffer ceilings used by CI systems.

I: AndroidSdkDriver: compile/test timeoutMillis = 600_000 (10 min), assemble timeoutMillis = 900_000 (15 min), maxOutputBytes = 4*1024*1024 (4 MB), maxOutputLines = 5000. Tasks: compileDebugSources --no-daemon --stacktrace, testDebugUnitTest --no-daemon, assembleDebug --no-daemon. APK path: app/build/outputs/apk/debug.

Foreseeable error: silent truncation; enforce the ceilings.

WIRE: AndroidSdkDriver, ToolchainExecutionDriver, BoundedProcessRunner

SD7-R-071+Kahn-token-floor [RESIDUAL]

Track: S

dependsOn: [SD7-T1-018]

R: Research minimum token-weight floors that prevent zero-cost atoms from collapsing partitions.

I: atom.estimatedTokens.coerceAtLeast(250). Apply inside Ψ . Foreseeable error: zero-weight atoms; clamp.

WIRE: PartitionClusterManager

SD7-R-072+chaos-error-rate-ceiling [RESIDUAL]

Track: B

dependsOn: [SD7-2050-FMC-02]

R: Research quantitative chaos acceptance thresholds used by production resilience systems.

I: During Digital Twin chaos: error rate must remain $< 10^{-6}$; zero unhandled promise or background thread rejections. Foreseeable error: qualitative 'survived'; require the numeric bound.

WIRE: Chaos Digital Twin, VerifiedCompletionGate, EvidenceCollector

SD7-R-073+partition-10k-node-benchmark [RESIDUAL]

Track: B

dependsOn: [SD7-T1-018]

R: Research synthetic DAG generators that produce 10 000-node acyclic graphs for partitioner validation.

I: ATOM-SYS-PL-001 verification: partitioner must pass against a synthetic 10 000-node dependency DAG with zero cycles. Foreseeable error: only testing small graphs; make the 10 k-node case a CI gate.

WIRE: PartitionClusterManager, CI job

SD7-R-074+ingestion-5k-page-512MB [RESIDUAL]

Track: B

dependsOn: [SD7-SG-001]

R: Research streaming ingestion under hard RAM ceilings.

I: ATOM-ING-001 verification: process a 5 000-page specification stream under a strict 512 MB RAM ceiling. Foreseeable error: materializing the whole stream; stream only.

WIRE: SpecGraphCompiler, CI job

SD7-R-075+schema-ephemeral-sqlite [RESIDUAL]

Track: B

dependsOn: [SD7-T4-001]

R: Research ephemeral in-memory SQLite validation of generated DDL and RLS before migration emission.

I: ATOM-L1-003 verification: run generated DDL/RLS against an ephemeral in-memory SQLite parser before emitting migrations. Foreseeable error: emitting unvalidated SQL; always validate first.

WIRE: StorageSchemaCompiler, SupabaseSchemaDriver, VerifiedCompletionGate

9. Additional Verification Domains, Governance & Formal Engines

SD7-R-080+WCAG-STRIDE-ILSS [RESIDUAL]

Track: S

dependsOn: [SD7-2050-ILSS-02]

R: Research automatic grafting of WCAG 2.2 AA accessibility contracts and STRIDE threat models from domain corpora.

I: ILSS DLOI extrapolation must include WCAG 2.2 AA and STRIDE as mandatory non-functional lattices when the domain implies UI or networked attack surface. Foreseeable error: optional; make them first-class Wave-0 atoms for relevant domains.

WIRE: ILSS, DloiService, ContractLatticeSynthesizer

SD7-R-081+Dafny-JavaPathfinder [RESIDUAL]

Track: B

dependsOn: [SD7-2050-NSV-01, SD7-2050-NSV-02]

R: Research integration of Dafny and Java Pathfinder as additional formal verification oracles alongside Kani/Creusot/icontract.

I: Neuro-symbolic SMT path must support Dafny and Java Pathfinder for languages/toolchains where they are the strongest available. Foreseeable error: single-oracle lock-in; keep the ensemble open.

WIRE: LanguageToolchainDriver, SMT verification path, VerifiedCompletionGate

SD7-R-082+sequential-Director-merge-arbitration [RESIDUAL]

Track: S

dependsOn: [SD7-T2-010, SD7-T1-040]

R: Research sequential merge arbitration locks that prevent concurrent worktree merges.

I: SUBAGENTS_NEVER_MERGE_DIRECTLY; ALL_MERGES_ARBITRATED_SEQUENTIALLY_BY_DIRECTOR_UNDER_VerifiedCompletionGate. Implement a Director-held merge lock. Foreseeable error: parallel merges; serialize.

WIRE: DirectorAgent, IsolatedWorktreeService, VerifiedCompletionGate

SD7-R-083+UTF8-key-decoder-ring-buffer [RESIDUAL]

Track: B

dependsOn: [SD7-T3-005]

R: Research searchable ring buffers and raw UTF-8 key decoders for terminal input.

I: CURSOR_KEYBOARD = VT100_ESCAPES + UTF8_RAW_KEY_DECODER + SEARCHABLE_RING_BUFFER. Complete the terminal input substrate. Foreseeable error: incomplete key handling; implement the full triple.

WIRE: CLI input loop, Terminal Input Defense

SD7-R-084+termux-clipboard-set [RESIDUAL]

Track: B

dependsOn: [SD7-T3-006]

R: Research Termux native clipboard binary binding.

I: OSC-52 fallback cascade must explicitly include termux-clipboard-set as the first Android/Termux target. Foreseeable error: missing Termux path; place it first in the cascade.

WIRE: OSC-52 clipboard bridge, SecretSinkMatrix

SD7-R-085+zero-reward-fraud-penalty-ledger [RESIDUAL]

Track: S

dependsOn: [SD7-T1-071]

R: Research zero-reward penalty ledgers that permanently record fraudulent synthesis attempts.

I: When AST stub scanner rejects code as fraudulent, log a ZERO_REWARD_PENALTY entry into the governance ledger. Future routing may down-rank the responsible provider. Foreseeable error: silent reject; always ledger.

WIRE: ArchitectureComplianceChecker, GovernanceLedger, RoutePolicy

SD7-R-086+true-moat-governance-ledger-amendment-chain [RESIDUAL]

Track: S

dependsOn: [SD7-T2-001, SD7-T2-025]

R: Research the true moat components: stateful evidence substrate, CAS-addressable records, restart continuity hashes, governance ledger, amendment chain.

I: TRUE_MOAT = STATEFUL_EVIDENCE_SUBSTRATE + CAS_ADDRESSABLE_RECORDS + RESTART_CONTINUITY_HASHES + GOVERNANCE_LEDGER + AMENDMENT_CHAIN. Implement GovernanceLedger and AmendmentChain as first-class durable stores. Foreseeable error: treating moat as rhetoric; wire every component.

WIRE: GovernanceLedger (new), AmendmentChain (new), CAS, EvidenceCollector, RestartCoordinator

10. Outside-the-Box Mathematical & Systems Advances

Atoms that push beyond Source Document 7 using techniques from formal methods, distributed systems, baseball/football analytics-style measurement, and CLI design that the original documents have not yet incorporated.

SD7-R-090+information-bottleneck-partitioning [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-018, SD7-R-051]

R: Research the Information Bottleneck method (Tishby) applied to graph partitioning so that each partition maximizes relevant mutual information with its contracts while minimizing leakage to sibling partitions.

I: Replace pure greedy domain-path clustering with an IB-inspired objective that explicitly minimizes $I(P_r; P_s)$ while preserving predictive power for the contracts.

Foreseeable error: intractable optimization; use variational approximation with hard territory constraints.

WIRE: PartitionClusterManager

SD7-R-091+differential-privacy-for-prompts [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-021]

R: Research differential-privacy noise injection into prompts that still preserve exact interface contracts (useful when multiple untrusted providers see the same Wave-0 lattice).

I: When a contract lattice is sent to more than one provider family, apply calibrated noise to non-semantic fields while keeping type signatures bit-identical. Foreseeable error: breaking typecheck; never noise the actual signatures.

WIRE: AgentPromptContract, ProviderCascadeRouter

SD7-R-092+sabermetrics-acceptance-velocity [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-R-057]

R: Research sabermetrics-style rate statistics (OPS, WAR analogues) for autonomous software factories that normalize AcceptanceVelocity by partition difficulty and provider cost.

I: Define $\text{FactoryWAR} = (\Delta \text{VerifiedPredicates} - \text{expected}) / \text{cost-normalized replacement level}$. Track per-provider and per-domain. Foreseeable error: vanity metrics; keep the definition anchored to Phase-100 predicates.

WIRE: DirectorAgent, GovernanceLedger, QuotaLedger

SD7-R-093+cli-progress-as-pure-stderr-protocol [AMENDMENT-CANDIDATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research pure-stderr progress protocols used by modern CLIs (cargo, npm, docker) that never contaminate stdout machine output.

I: All progress, spinners, and human telemetry must be pure stderr with an explicit --json-progress alternative that emits NDJSON on stderr only. Stdout remains pure artifacts or machine JSON. Foreseeable error: progress on stdout; forbid by architecture test.

WIRE: CLI TTY discipline, all diagnostic emitters

SD7-R-094+formal-proof-carrying-code-bundle [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-R-052, SD7-2050-LHD-04]

R: Research proof-carrying code (Necula) and modern PCC variants that ship machine-checkable proofs alongside binaries so that a third party can validate without re-running the LLM.

I: Extend the Zero-Entropy Proof Certificate into a full proof-carrying bundle: SpecHash + NegativeSpecHash + SMT proofs + TLA receipts + mutation-kill score + chaos receipt + toolchain binary hash + optional PCC certificate. Foreseeable error: missing fields; VerifiedCompletionGate refuses promotion until complete.

WIRE: Zero-Entropy Proof Certificate, IntegrityProofs, VerifiedCompletionGate

SD7-R-095+online-convex-optimization-for-routing [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-024, SD7-ADV-004]

R: Research online convex optimization / multi-armed bandit algorithms that continuously re-rank providers under non-stationary quality and cost.

I: Replace static RoutePolicy rankings with an OCO/bandit update after every verified atom. Regret bounds must be logged. Foreseeable error: catastrophic forgetting; keep a slow-moving prior.

11. Final Residual Completeness Gate

SD7-R-099+residual-completeness-gate [RESIDUAL]

Track: S

dependsOn: [ALL residual atoms above]

R: Research orphan and residual-atom detection that proves every file, constant, equation and invariant named in Source Document 7 and the SD6 addenda now has a corresponding atom.

I: atropos doctor --residual-completeness must pass. It cross-checks the three DAG documents against the full SD7 + SD6 inventory and fails on any remaining omission.

Foreseeable error: false green from partial matches; require exact file-path and equation coverage.

WIRE: CI job, atropos doctor, ArchitectureComplianceChecker

12. Ordering Relative to Prior DAGs

Tier-1 Hybrid DAG (template purge + Wave factory + 2050 neuro-symbolic) must be green first.

Full Multi-Tier DAG (Tiers 2/3/4 + SpecGraph expansions + advanced research) must be green second.

This Residual DAG is the third and final layer. SD7-R-099 is the absolute last gate before the system is declared complete with respect to Source Document 7 and its addenda.

13. Non-Negotiable Invariants (Restated for Residual Layer)

Entropy(C | S, I) = 0

$I(f_j; f_k | I) = 0 \quad \forall (v_j, v_k) \notin E_A$

$X(\tau_i) = I \cup \{a_i\} \cup \text{Ancestors}(a_i, I)$

$\sigma_{\{t+1\}} = \delta(\sigma_t, a_{\{t,t\}}, I)$

$\prod v(\Phi(P_r, I)) = 1$

$\Delta f \subseteq \text{Territory}(P_r)$

$V_{\text{accepted}} = \min(V_{\text{generated}}, V_{\text{integrated}}, V_{\text{verified}}, V_{\text{evidenced}})$

Phase 100 full conjunction

False-Green Tolerance = 0.0000 %

Air-Gap for pure synthesis

Exact-hash restart continuity

END OF RESIDUAL OBLIGATION DAG — Final Missed-Atom Inventory

Third and final document in the Source Doc 7 series · Leaves off from Tier-1 + Multi-Tier DAGs · Byte-level completeness · Outside-the-box advances included · 2026-10-06